

Switch Loop Detection by Machine Learning

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ABSTRACT

The switch loop detection system by supervised machine learning is built to verify the network status whether the system is in normal condition or there are loop occurs within the system. There are several attributes that have been selected from the switch configuration along with the different categories of supervised machine learning were used to determine the best model successful rate. In this paper, it shows that Decision tree (DT) has higher accuracy rate compared to support vector machine (SVM) model. Moreover, we are trying to illustrate the course and effect of this issue by creating the graphic user interface (GUI) and monitor it through web browser.

I. INTRODUCTION

Nowadays, Networking become one of the most important facilities which is use in many situations by an individual at all time. The internet has been used as a part of many application such as in China which has turned into a networked country because of rapidly development in internet of Things, Internet, big data and cloud computing which is importance to people and has continuously turn into part of human life due to its high efficiency, specificity and cross-domain characteristics [1]. However, there are some situations that create a loop which result to destroy and shutdown the whole network system and one of the reasons is switch loop which might occur in a different situation. When the loop occurs, the exchange of information will be affected especially the application that use real time communication such as internet of things and cloud computing will surely suffer from bad consequences.

One of the well-known network detection system is Intrusion Detection Systems (IDS). It was established in 1980 to provide network security based on three principles: confidentiality, integrity, and privacy. The first model was introduced and published in 1987 [2]. Since it was introduced, it became famous topic and has been developing to

improve its efficiency. Intrusion Detection Systems (IDS) can be categorized into two categories which are Network-based Intrusion Detection System (NIDS) that monitor incoming and outgoing traffic of all devices which focused on network level and Host-based Intrusion Detection System (HIDS) which focus on entire network including information from devices log which determine the change within the system

II. BACKGROUND AND RELATED WORK

A. Spanning Tree protocol (STP)

It is a layer 2 protocol which runs on network switches. The main function of this protocol is to prevent loop occur when there is redundant path inside the system. There are 4 main device port states when STP is enable which are 1) Blocking state to block all incoming packets to prevent loop path. 2) Listening state to wait for upcoming information from the neighbor devices. 3) Learning state to learn and remember the source address of the packet and 4) Forwarding state when ports are ready to receive and send out the packet. After it reaches the final state, the network should be stable and allow all devices send and receive data from one another.

B. Ansible

It is known as the open-source program which used for device configuration, and deployment. It was founded in 2013 and was acquired by Red Hat in October 2015. The architecture of ansible is depends on python and the controlled devices must have python and ansible package. The connection between ansible and devices is occurred via Secure Shell (SSH) or PowerShell execution in Windows Remote Management. The advantages of ansible are that it is able to communicate many devices and able to deploy the configuration to all of them at a time by a single command execution. Also, it is secure and reliable because of the connection using Secure Shell (SSH) [3].

C. Network and Machine Learning research

There are many researches and knowledge about machine learning which has adapted into real applications and used for network application. One of the researches of this application is Network Intrusion Detection using Supervised Machine Learning Technique with Feature Selection which was published in 2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST) [4]. The researchers manage to use different type of supervised machine learning to differentiate and analyze the network traffic and select Artificial Neural Network (ANN) as a best model with the highest accuracy.

III. SYSTEM MODEL

The system composed of data retrieval from physical devices and machine learning algorithm. We used ansible to communicate and retrieved data from the switches. The data will be stored in SQL database. The data set will be split into training and testing set by using cross validation function to ensure that every record will be used to train and test the algorithm. After the models are tested, their performance will be compared with one another to selected the best model and will be used to classify whether there is no loop or what might create loop in the system so that the responsible person can respond accordingly.

A. Database Creation and Data retrieval

The initial state of doing this research is to prepare the environment including the database system to store the retrieved data which will be used to created model. We use SQL to create database and tables. There are the table that stored raw data and the table stored the aggregated record. Moreover, we have created the table for the device connection or topology which allowed us to understand the connection among the network device where each of them is representing as a node that connecting among one another by the physical connection through their physical port [5]. Then, we will use this information to plot the topology chart which will be shown on the webpage.

After we create the database, we tried to create several environments in the system and collected data simultaneously by writing the script using ansible along with SQL command to store data in database. During the data collecting phase, the physical devices configuration has been changed and enormous number of packets will be sent into the system.

The script will be called and running repeatedly by crontab function in Linux command. When raw data is added into table, we use some aggregate function to summarize the condition of system and store it into another table so that these records will be used to create model later. The reason behind the aggregation of data and store them into another table is to optimize and increased the performance of query when there is multiple access to the database.[6]. However, there are some situations that physical switch could not respond so the raw data might be record manually.

B. Decision tree model

The decision tree can be described as one of the most intelligible algorithms which is known globally. It can be constructed by two components which are node and branch. The first top node is called decision node and followed by chance node and end nodes which is the output predicted from the algorithm. During the training process, we have to select one attribute as a class and the model will choose an attribute that can split data into subset. It started to calculate the entropy of the attributes. The formula to find an entropy is described below.

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i \quad (1)$$

After the entropy is calculated, it will proceed to compute the information gain and the attribute with the most information gain will be selected as a decision node. The formula which uses to compute information gain is shown below.

$$Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) \quad (2)$$

If the decision node is not able classify all record into different types of classes, then the algorithm will start to find other nodes until it can classify all record.

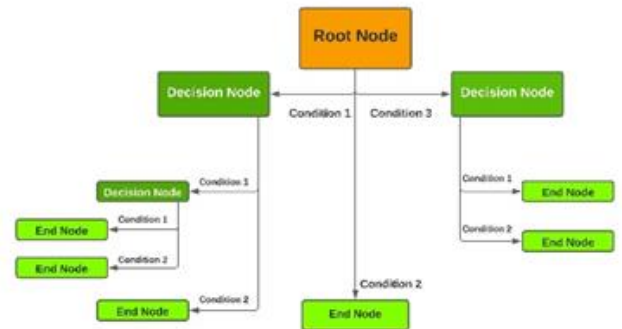


Fig1. Decision Tree Model

C. Neuron Network model

Neuron network is known as system which stimulate living organism's brain. It relies on the connection of nodes in the system and signal or information transmission to edges. There is weight to modify the learning process which used to control the power of the signal. It might consist of threshold that allow aggregated signal transfers though it. Moreover, in common neuron network, the nodes will be composited into layers. The calculating process will be initialized at the transmission in first layer until the last layer to predict the outcome. The node and hidden layer should be optimized when train the model because of increasing them might not improve the performance but may create the complexity to the model [7],[8]. The example of neuron network topology is represented below.

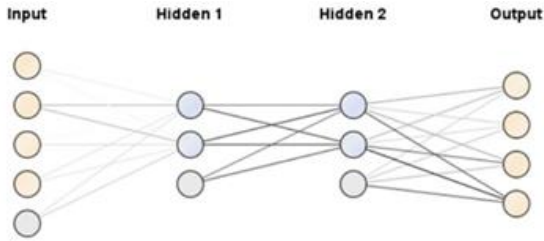


Fig2. Neuron Network with its hidden layer

D. Support Vector Machine model

Support Vector Machine (SVM) is a classification model which is used for machine learning. The algorithm aims to classify the data by creating the line that can separate data into different class. This line is described as a hyper plane [9],[10]. The closet data of each output class to the hyper plane is known as support vector and the distances between them is called margin. However, the support vector machine can be classified by the use of its kernel or a set of mathematical function to transform the input data into a specific form to create a model. There are several kernels which commonly use such as linear, nonlinear, polynomial, radial basis function (RBF), and sigmoid.

IV. Result Comparison

After we have retrieved data in several environments and store them into specific database table as records. Those records were used to train and test to compare the performance for each model. All records are using as a training and testing dataset by the technique of cross-validation [11]. We have used a total of 10086 records and split it into 10

cross-validation sets to train and test the model. It appeared that Decision Tree model has outperformed in both training time taken and the accuracy from other algorithms. The performance has shown in the table below.

Machine Learning Algorithm	Training Time	Accuracy
Decision Tree	2 seconds	95.77%
Support Vector Machine	6 seconds	93.57%
Neuron Network	105 seconds	91.24%

Table1: The Performance of each algorithm

V. Implementation

This research could be adopted and used in many situations. To illustrate, we have created web application with the use of machine learning model. As Decision Tree has been selected as an outperform model in this research, it could be used to classify and detect the network system failure weather the network is stable or there is loop occurred. The process started with data retrieval from devices using ansible. The retrieval data are separate into 2 table which are the devices connection and the status of network system. The device connection information is used to created visual topology on the webpage and the network status records are used for analyze the status via machine learning model.

After the model has predicted the outcome, it will record the status into a specific database table. Then, webpage will create topology and present the predict status on the screen by the communication via PHP with database. The described process is shown in the diagram below.

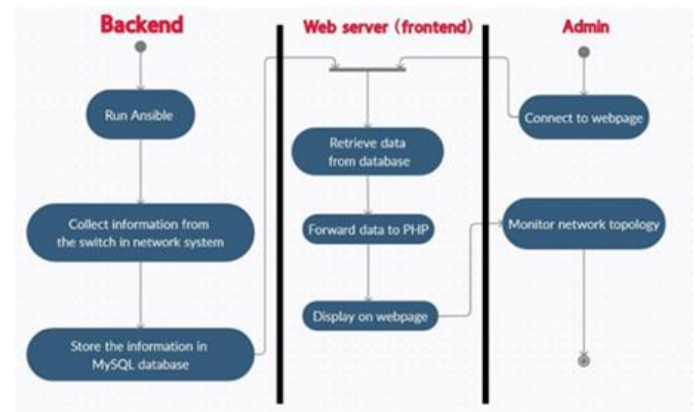


Fig3. The process of application

The topology is shown on interface by using Anychart which is JavaScript library for creating the dashboard. Each node on the board represents one device from the system and the line connected between them is represent the physical connection of switches. The object information will be presented when user move the mouse cursor over the object on the dashboard.



Fig4. The topology present on webpage during normal situation

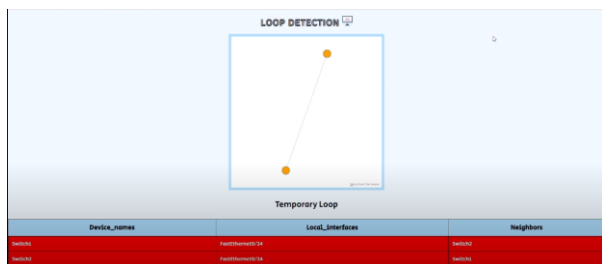


Fig5. The topology present on webpage when loop occurred

VI. Suggestion

We believe that this research could be implemented in several project and future work. However, it could be developed by increasing the attribute of data or test in different environment with the different hardware product. However, this must be concern due to the increasement of time during data retrieval period. Moreover, when loop occurred, the CPUs load will be increased rapidly which might result in incomplete data retrieval and the model might not be able to process in real-time operation. Therefore, this research can be contributed and develop in the future work.

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